

Rik L.E.M. Ubaghs

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Bio

I am an engineer specializing in neurotechnology and optical systems, with expertise in biomedical imaging, medical devices, and data analytics. My work focuses on translating advanced experimental technologies into practical clinical applications, bridging the gap between cutting-edge research and real-world medical impact.

Education

ETH Zurich, Zurich, Switzerland <i>Ph.D. Neurotechnology</i>	July 2017 - May 2023
University of Amsterdam, Amsterdam, the Netherlands <i>Research Master Brain and Cognitive Sciences, Cum Laude</i>	Sept 2014 - June 2016
Maastricht University, Maastricht, the Netherlands <i>Bachelor Psychology and Neuroscience, Cum Laude</i>	Sept 2011 - June 2014

Skills

Languages:	Dutch (Native), English (Fluent), German (Basic)
Programming:	Python, Rust, Matlab, R
Databases/Querying:	SQL, MySQL
Skills:	Data Science, Biomedical Imaging, Optical Systems, Mechanical Design (Autodesk, Solidworks)

Industry Positions

Arago Labs <i>Co-founder, CEO</i> - Founded an early-stage medical technology venture focused on improving surgical precision in oncology. - Designed advanced optical imaging platform combining nonlinear microscopy, and ML/DL-based tissue classification. - Defined market entry strategy, pricing model, reimbursement pathways, and go-to-market roadmap.	June 2023 - Current
Agora Technology, Maastricht, The Netherlands <i>Founder</i> - Built data solutions that aimed at streamlining/automating the workflow of medical facilities, especially focusing on surgical equipment. - We focused on internal logistics, and the creation, maintenance, and analysis of large-scale medical equipment databases using ML/DL.	Nov 2016 - Feb 2018

Research Positions

Inst. of Neuroinformatics, ETH Zurich, Zurich, Switzerland

July 2017 – June 2023

Ph.D. Candidate

- Development of a MRI compatible fluorescence microscope (full-cycle product design).
- Design of a paradigm to combine microscopy and MR imaging methods in awake rodents.
- Design of data pipelines for multi-modal data analysis.

Center for Neuroengineering, Duke University, Durham, USA

Dec 2015 – Nov 2016

Visiting Scientist

- Design of a hybrid brain decoding strategy in Non-human Primates.
- Development of ML/DL decoding pipeline for brain decoding.

Spinoza Center for Neuroimaging, Amsterdam, The Netherlands

Nov 2014 - Dec 2015

Junior Research Scientist

- MR operator.
 - Design and supervision of several large-scale research projects.
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Extracurricular Positions

Natural Movements, The Netherlands

March 2007 - May 2012

Co-Founder

- Natural Movements is a non-profit organization that aims to engage underprivileged youth in the local community through behavioral coaching and physical education.
 - Worked together with social service organisations to provide youth with any additional support they required.
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Publications

Publications: - Simultaneous single-cell calcium imaging of neuronal population activity and brain-wide BOLD fMRI Ubaghs, R.L.E.M., Boehringer, R., Marks, M., Hesse, H.K., Yanik, M.F., Zerbi, V., Grewe, B.F., bioRxiv 2023.11.14.566368; doi: <https://doi.org/10.1101/2023.11.14.566368>; Accepted Nature Methods

Conference Papers:

- Ubaghs, R.L.E.M., Böhringer, R., Marks, M., Yanik, M.F. and Grewe, B.F., The Effect of Somatostatin+ Interneurons on the Negative BOLD Response. Proc. Intl. Soc. Mag. Reson. Med. 29 (2021)
- Ubaghs, R.L.E.M., Dermutz, H., Böhringer, R., Marks, M., Yanik, M.F. and Grewe, B.F., Shaping the BOLD signal through Excitatory and Inhibitory Interaction. Proc. Intl. Soc. Mag. Reson. Med. 28 (2020)

Patents:

Sainz Villalba, L., Ubaghs, R. DEVICE AND METHOD FOR PROVIDING HISTOLOGICAL NAVIGATION INFORMATION DURING IN-VIVO SURGERY. European Patent Application, Filed 2026.

Honors

Fellowships:

Pioneer Fellow, ETH Zurich

April 2025 - Current

Committees:

Educational Committee, IIS, University of Amsterdam

Sept 2014 - June 2015

Vice-President Student Council, FPN, Maastricht University

Sept 2012 - June 2013

Program Committee, FPN, Maastricht University

Sept 2012 - June 2013